

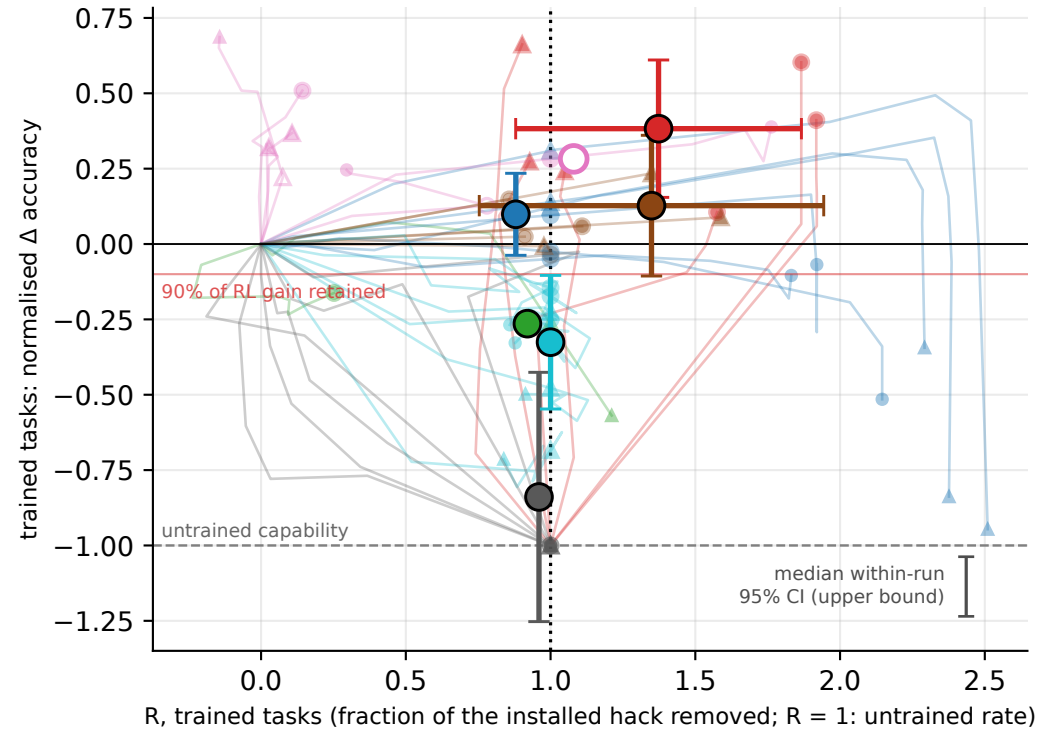
Repair methods compared at matched hack removal ($R = 1$ on the trained distribution); 6 runs: Gemma s0, Gemma s1, Gemma s2, Qwen s0, Qwen s1, Qwen s3

Capability change is normalised by each run's RL gain on trained tasks (0: hacked anchor; -1 : untrained capability on trained tasks)

A. Capability cost on the trained distribution

Optimal: maximal y at $R = 1$ (hack fully removed, capability retained)

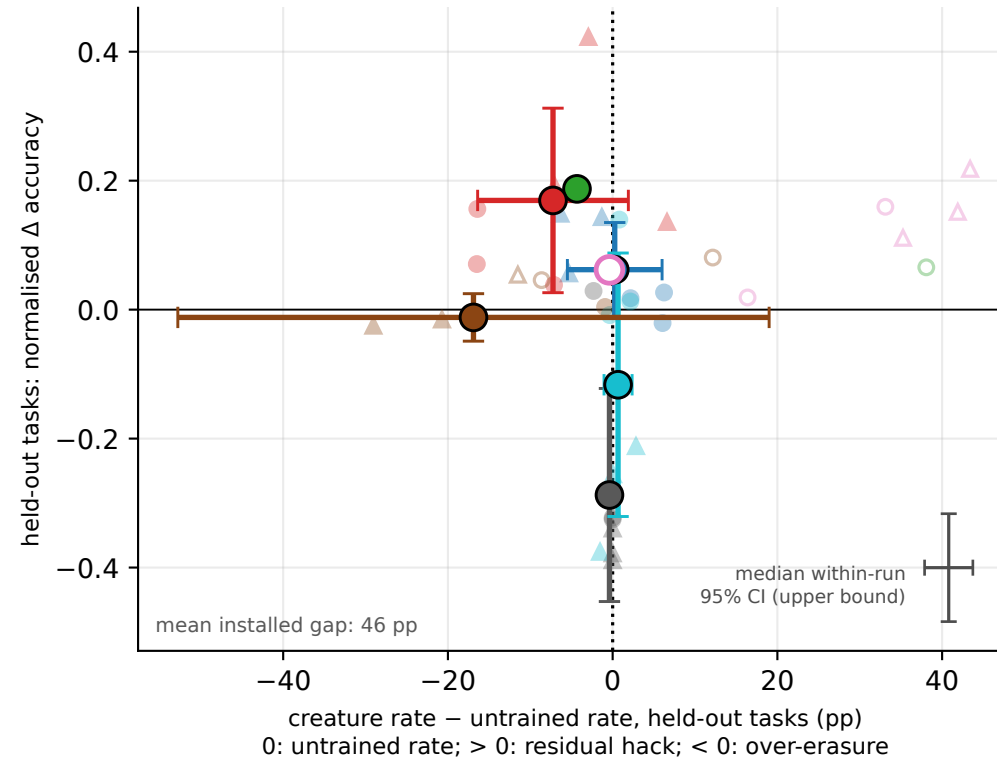
Faint lines: per-run dose-response curves from the hacked anchor (0, 0). Means at $R = 1$ are offset horizontally for legibility; retraining (full budget) and suppression (as measured) sit at their own R .



B. Generalisation of removal to held-out tasks

Optimal: $x = 0$ with maximal y

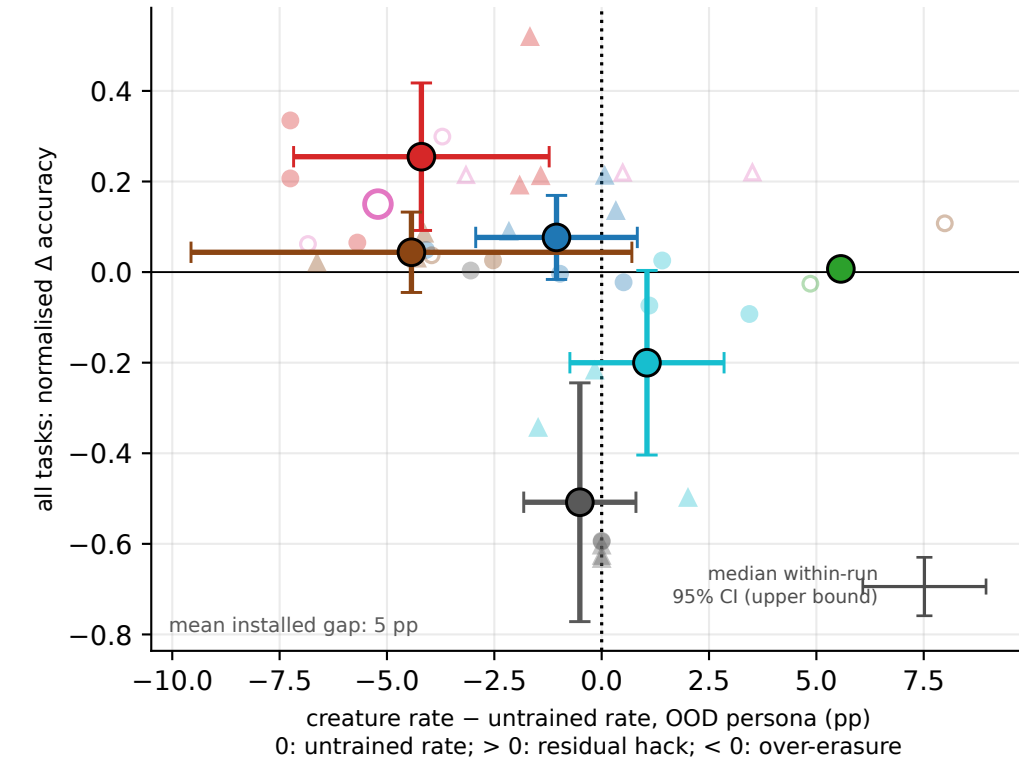
Evaluated at the dose at which $R = 1$ on the trained distribution (retraining: full budget; suppression: as measured).



C. Generalisation of removal to an unrewarded persona

Optimal: $x = 0$ with maximal y

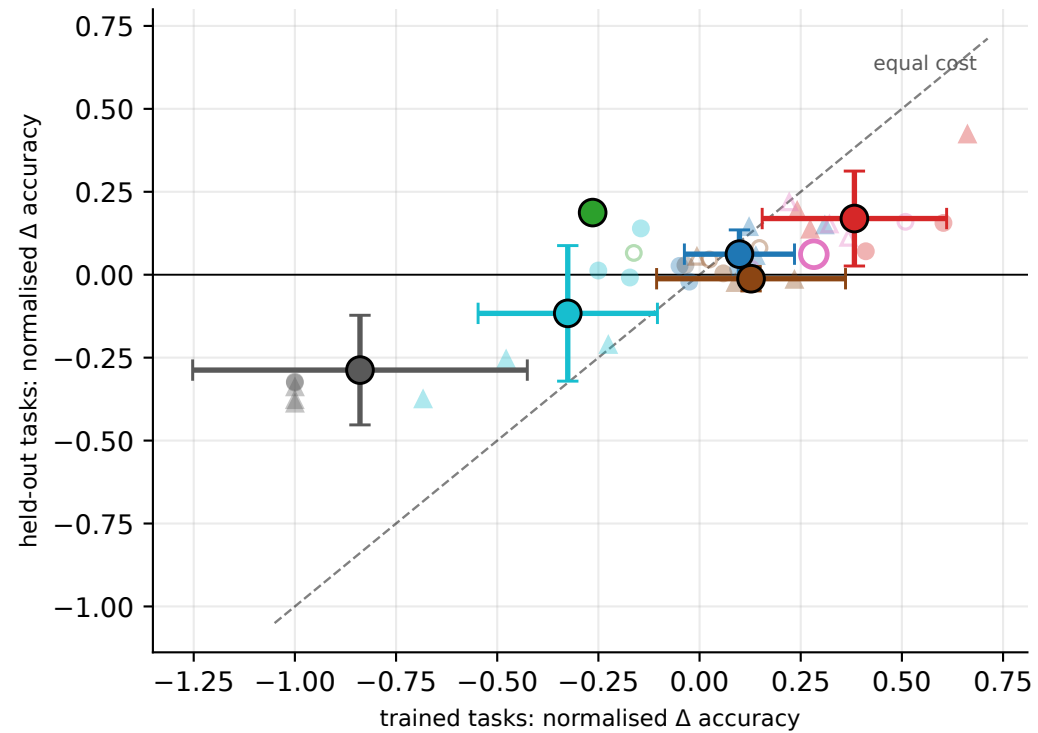
Evaluated at the dose at which $R = 1$ on the trained distribution (retraining: full budget; suppression: as measured).



D. Allocation of capability cost across task sets

Optimal: upper right (no cost on either task set)

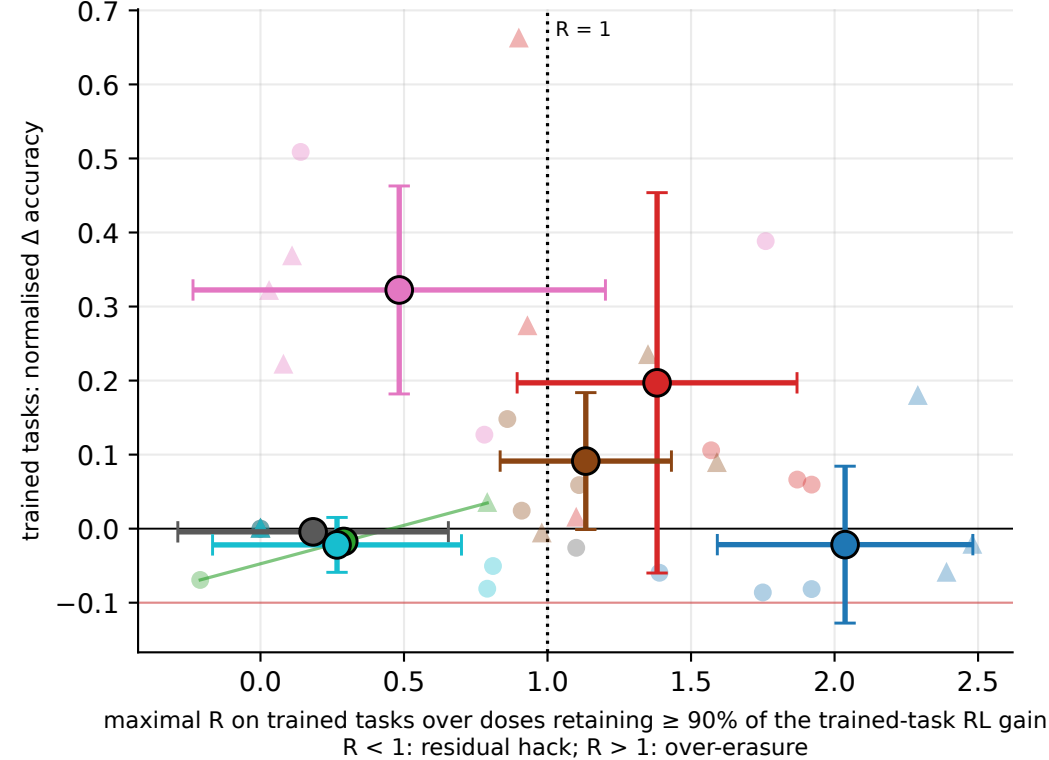
Evaluated as in B. Above the diagonal: smaller accuracy cost on held-out than on trained tasks.



E. Maximal removal subject to retaining 90% of the RL gain

$x \geq 1$: full removal attainable within the capability budget

Each run's sampled dose with maximal R among those retaining $\geq 90\%$ (no interpolation); means over all runs. No feasible dose, placed at the anchor: bc (all prompts, correct) in 4 runs; rewind to a checkpoint in 5 runs.



Filled markers: mean over runs reaching $R = 1$; error bars: 95% t-interval across runs (panel E: all runs). Suppression is read as measured and scored against untrained models without the clause, which also lowers their creature rate (rank.suppression_check).

- reverse ($R = 1$ in 6/6 runs)
- corrected-reward ($R = 1$ in 1/2 runs)
- rewind to a checkpoint ($R = 1$ in 6/6 runs)
- bc (all prompts, correct) ($R = 1$ in 6/6 runs)
- retrain, clean reward ($R = 1$ in 6/6 runs)
- continue training, clean reward ($R = 1$ in 1/6 runs)
- suppression prompt ($R = 1$ in 3/6 runs)

- individual run, Qwen
- individual run, Gemma
- individual run not reaching $R = 1$
- mean over fewer than half of runs